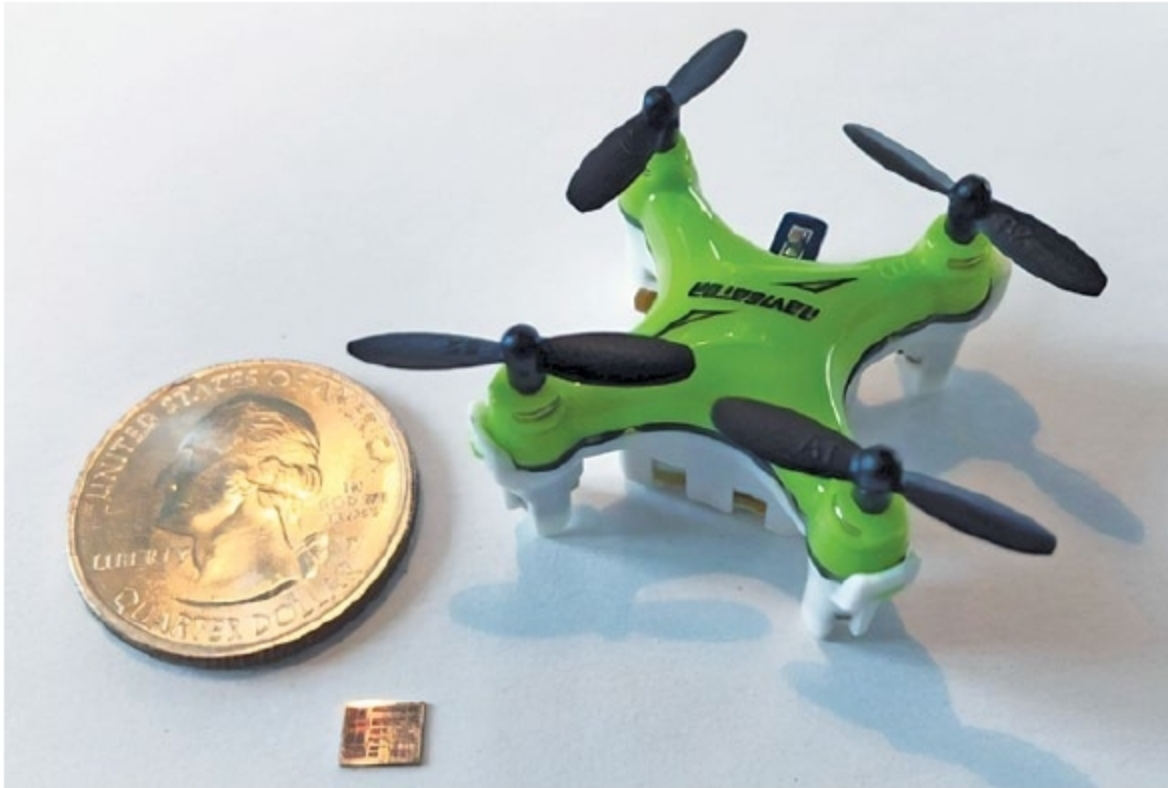


Chip upgrade to help miniature drones navigate



The computer chip, which is smaller than a US dime (shown here with a quarter for scale) helps miniature drones navigate in flight (Credit: MIT)

Last year, researchers at Massachusetts Institute of Technology (MIT) designed a tiny computer chip tailored to help honeybee-sized drones navigate. They have now shrunk the chip design even further, in both size and power consumption.

where GPS data is unavailable.

The chip design can also be run on any small robot or device that needs to navigate over long stretches of time on a limited power supply.

The team, co-led by Vivienne Sze, associate professor in MIT's Department of Electrical Engineering and Computer Science, and Sertac Karaman, class of 1948 - career development associate professor of aeronautics and astronautics, built a fully-customized chip from the ground up, with a focus on reducing power consumption and size, while also increasing processing speed.

The new computer chip, called Navion, is only 20 square-millimetres—about the size of a LEGO mini-figure's footprint—and consumes just 24 milliwatts of power.

Using this tiny amount of power, the chip can process camera images in real time, at up to 171 frames per seconds as well as inertial measurements, both of which it uses to determine where it is in space. According to the researchers, the chip can be integrated into nano-drones as small as fingernails, to help vehicles navigate, particularly in remote or inaccessible places

Robots, prosthetic devices get a sense of touch

Stanford University and Seoul National University researchers have developed an artificial sensory nerve system that can activate the twitch reflex in a cockroach and identify letters in Braille alphabet. This is a step towards creating artificial skin for prosthetic limbs, to restore sensation in amputees and, perhaps, one day give robots some type of reflex capability.

Zhenan Bao, a professor of chemical engineering, says, "We take skin for granted, but it is a complex sensing, signalling and decision-making system." This artificial sensory nerve system can make skin-like sensory neural networks for all sorts of applications.

This breakthrough is part of Bao's quest to mimic how skin can stretch, repair itself and, most remarkably, act like a smart sensory network that knows not only how to transmit pleasant sensations to the

brain, but also when to order the muscles to react reflexively to make prompt decisions.



An artificial nerve system developed at Stanford gives prosthetic devices and robots a sense of touch (Credit: Stanford University)